MIL-C-46357(MU)
AMENDMENT 5
28 April 1969
SUPERSEDING
AMENDMENT 4
16 October 1967

MILITARY SPECIFICATION

COMPUTER, GUN DIRECTION: M18
MANUFACTURING PROCESSES AND INSPECTION OF

This amendment forms a part of Military Specification MIL-C-46357(MU), dated 19 March 1965.

Paragraph 2.1: Delete the following Military Specifications:

"MIL-S-6872	Soldering Process, General Specification for
MIL-P-18177	Plastic Sheet, Laminated, Thermosetting, Glass Fiber Base, Epoxy-Resin"

Paragraph 2.1: Add the following Military Specifications:

"MIL-T-10727	Tin Plating, Electrodeposited or Hot-dipped, for Ferrous or Non-ferrous Metals
MIL-S-45743	Soldering Electrical Connections, for Guided and Ballistic Missiles, with Electrically-Heated Soldering Irons, and Resistance Soldering Apparatus; Procedures for"

Paragraph 3.2.1: Delete in its entirety and substitute the following:

"3.2.1 Circuit board materials.- Unless otherwise specified the copper clad plastic sheet shall conform to MIL-P-13949, Type FL-GE-0.062-C-2/2-CLIII or Type FL-GE-01062-C-2/0-CLIII as required by applicable drawings."

Paragraph 3.2.3.2(c) Delete and substitute:

"(c) Tin-Lead plating.- There shall be a minimum of 0.0003 inch tin-lead plating on the conductor pattern if tin-lead plating is deposited over the copper. The tin-lead plating shall contain a minimum of 50 percent and a maximum of 70 percent tin. Tin-lead plating shall meet Specification MIL-T-10727."

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Add subparagraph (d)

"(d) Board contacts.- Board contacts in the 0.312 inch engagement area, for tin-lead and gold plated boards shall be gold plated in accordance with either a. or b. above. The roughness finish of the plated contacts shall not exceed 32 micro inches RMS for any of the (3) types of plating. The thickness across contacts shall not exceed 0.0657 inches after plating. This is also applicable to circuit boards with copper cladding on one side."

Paragraph 3.2.3.3.1.- Third sentence, delete and substitute: "The walls of the plated-through holes shall be plated with uniform and continuous copper and gold; copper, gold and nickel; or copper, tin-lead plate, without breaks in either the copper, gold nickel or tin-lead plating as applicable. Voids in excess of 10 percent of surface in a hole is cause for rejection."

Paragraph 3.2.3.4: Add to last sentence "nor more than .025 wide".

Paragraph 3.2.3.5(g): Delete in its entirety and substitute the following:

"(g) Copper or nickel visible through the gold; gold-nickel or tin-lead plating."

Add paragraph 3.2.3.5.2:

"3.2.3.5.2 Location grid.- The location grid shall be 0.050 inches basic increment. All board edges, indexing holes, part mounting holes, and cutouts shall be located on this grid unless otherwise located on applicable drawing. The index holes shall be within 0.003 inch radius of the true grid intersection and component mounting holes shall be within 0.005 inch of the true grid intersection."

Paragraph 3.2.5.1: Change first sentence to read as follows:

"The minimum unformed lead length between the component body and the lead bend radius (see 3.2.5.2) shall be 0.062 inch and shall be parallel to the circuit board component mounting surface."

Paragraph 3.2.5.2 add the following: "The component lead shall be bent at 90° ± 5° at a point that will result in verticle entry of the lead into the respective printed circuit board hole."

Paragraph 3.2.5.3: Delete and substitute:

"3.2.5.3 Clinching of leads.- All wire leads which are inserted into printed circuit boards shall be clinched on the side opposite the point of insertion. The length of the clinched end shall be such that no part of the lead shall be closer to a non-connected circuit path

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than the minimum spacing maintained between non-connected circuit paths on the board. The clinch shall be such that it holds the component part securely to the board so as to permit a good solderable contact."

Paragraph 3.2.6: Delete "3.2.6 Soldering and tinning" and substitute the following:

"3.2.6 Soldering process.- Unless otherwise specified herein, the soldering process shall conform to MIL-S-45743. Should any conflict exist between MIL-S-45743 and this specification, this specification shall take precedence."

Paragraph 3.2.6.1: Delete last sentence and substitute the following: "The solder used shall conform to MIL-S-571, Type SN60 or SN63."

Paragraph 3.2.6.3: Delete fifth sentence: "The edge of the maskant shall be applied along a line formed by prolonging the edges of the end notches across the board to within 1/64 inch of the line." Substitute the following: "The maskant shall be applied to the connector contacts with the edge of the maskant being 1/64 inch (in-board) above the silk screened numbers, except in areas where component leads interfere, in which event the maskant shall be removed from these areas only."

Paragraph 3.2.8.2: Revise next to last sentence to read:

"Except for areas marked, all surfaces shall be covered with a coating of .001 inch minimum to .005 maximum thickness. For exceptions to coating see 3.2.8.4."

TABLE I: Delete the first four lines of the columns and substitute the following new line:

"101.	Circuit board materials	3.2.1	MIL-P-13949"

Add the following:

"107.	Plating (Tin-Lead)	3.2.3.2(c)(d)	MIL-T-10727, VIS & SME
112a.	Grid location	3.2.3.5.2	VIS & SME"

Change the following to read:

"108.	Plating (Gold)	3.2.3.2(a)(d)	MIL-G-45204, VIS & SME
109.	Plating (Gold-Nickel)	3.2.3.2(b)(d)	QQ-N0290, VIS & SME

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TABLE I, under METHOD OF INSPECTION: Delete "MIL-S-6872" and substitute "MIL-S-571".

TABLE II, under METHOD OF INSPECTION: Delete "QQ-S571" and substitute "MIL-S-571".

TABLE VIII: Delete the following lines:

"301.	Circuit board material (copper	3.2.1(b)
304.	Solder Type SN60 or SN63 (QQ-S-571)	3.2.6.1"

and substitute the following new lines:

"301.	Circuit board materials	3.2.1
304.	Solder Type SN60 or SN63 (MIL-S-571)	3.2.6.1"

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